H

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION II
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME

CLASS | INDEX NUMBER

BIOLOGY 8875/02

Paper 2 Core Paper

17 September 2013

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **any one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**13**
2	**10**
3	**8**
4	**9**
Section B	
5 / 6	**20**
Total	**60**
Percentage	
Grade	

This document consists of **12** printed pages.

Section A

Answer **all** questions.

1 Blackflies are aggressive blood-feeders on people, domestic livestock and a variety of warm-blooded wildlife. Its larvae live in rivers and feed by filtering particles out of the water.

A method of controlling blackflies is to add particles of an inactive proteinous toxin to the water upstream of where the larvae are feeding. The toxin will be activated by a protease secreted by the stomach of the larva.

The active toxin then binds to a receptor molecule on the membranes of the gut epithelial cell. This damages the epithelial cells and kills the larva.

Fig. 1.1 is an electron micrograph of a stomach cell of the blackfly larva.

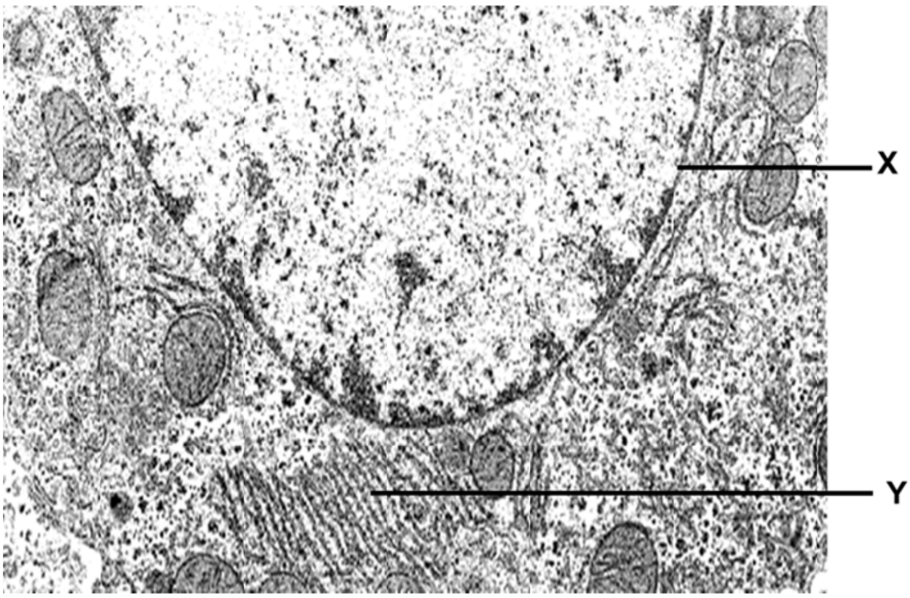

Fig. 1.1

(a) (i) Describe how organelle X is adapted for genetic expression.

1. Nuclear pores with proteins controlling movement of RNA and proteins between nucleus and cytoplasm;; (MUST)

2. Nuclear envelope separates nuclear compartment from cytoplasm (in eukaryotes) which allows spatial separation of the transcription and translation process → creates micro-environment that allows for post-transcriptional level of control / modification of transcripts before translation;;

3. Contain genes which code for products involved in genetic expression (eg. HATs);;

4. Contain enzymes e.g. HATs that chemically modify chromatin, affecting compactness and accessibility of promoter to transcription factors and RNA polymerase, hence allowing for genomic level of control of gene expression;; (KIV 3-4) [2]

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(ii) Describe how insulin, after being synthesized by ribosomes on Y, is exported out of the cell.

1. ***protein travels thru cisternae to bud off in a transport vesicle at transitional ER, transport vesicle travels through cytoplasm & fuse with GA at cis face;;***

2. ***protein enters cisternal space & modified by adding/deleting/substituting of sugar monomers in oligosaccharide chains &/or adding phosphate groups;;***

3. ***modified & tagged protein travels from cisternae to cisternae via budding & fusion of Golgi vesicle, sorted protein leaves GA in a secretory vesicle that buds off at trans face;;***

4. ***secretory vesicle moves near cell surface membrane via cytoskeleton, membrane of secretory vesicle fuses with plasma membrane → secretory protein released into exterior of cell via exocytosis;;*** [4]

Fig. 1.2 shows the results of an experiment to determine the optimum pH for activation of the toxin by the stomach protease.

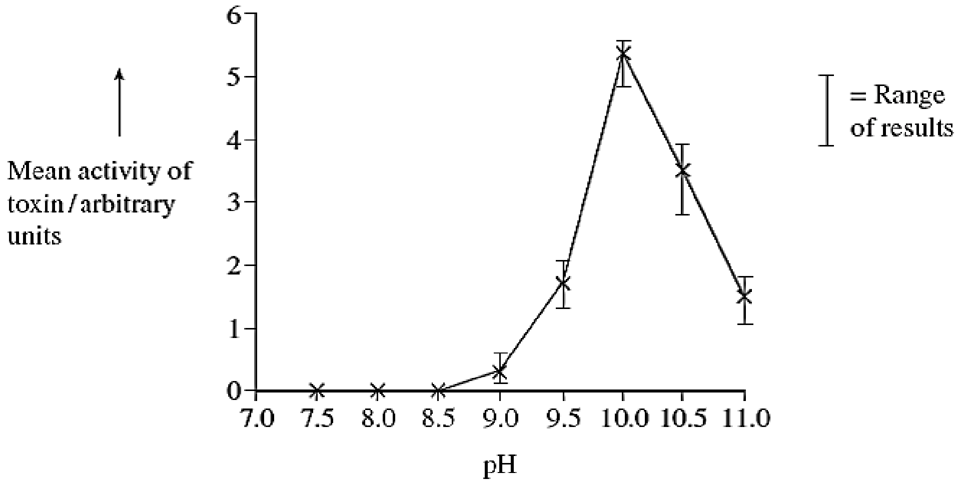

Fig. 1.2

(b) With reference to Fig. 1.2, describe and explain the effect of pH on the mean activity of the toxin.

1. ***Optimum pH at 10 at which there's highest mean activity of toxin between 5-6 a.u (specify one value) because rate of E-S complex formation is max;;***

2. ***either an increase in pH from 10 to 11 or decrease in pH from 10 to 9.5 results in drastic decrease of mean activity to 2 a.u (or less);;***

3. ***change in pH results in increase H+ concentration disrupts intramolecular ionic bonds and hydrogen bonds that help maintain the specific 3D conformation of the protease ® toxin →*** [3]

For Examiner's Use

denatured → no E-S complex formed → no activation of toxin;;

(c) Suggest how the protease in the larva's digestive system activates the toxin.

1. protease cleaves toxin when catalytic aa at its active site hydrolyses peptide bonds in toxin;;

2. toxin molecule acquires complementary configuration for binding receptor molecule/ exposes binding site for receptor molecule on gut epithelial cell membrane;; [2]

(d) The use of this toxin has been shown to be effective in eliminating the blackfly larvae population while having no adverse effect on other organisms living in the river.

Suggest **two** reasons why the toxin can be used for selective biological control of the blackfly.

1. Receptor molecule in gut epithelial cell to which toxin binds is found only in blackfly;;

2. Specific protease needed for activation of toxin found only in blackfly;;

3. Gut pH needed for activation of toxin found only in blackfly/ other organisms have different gut pH which does not activate protease which cleaves toxin to its active form;;

(any 2)

[2]

[Total: 13]

For Examiner's Use

2 Fig. 2.1 shows the light dependent reactions occurring in the chloroplast.

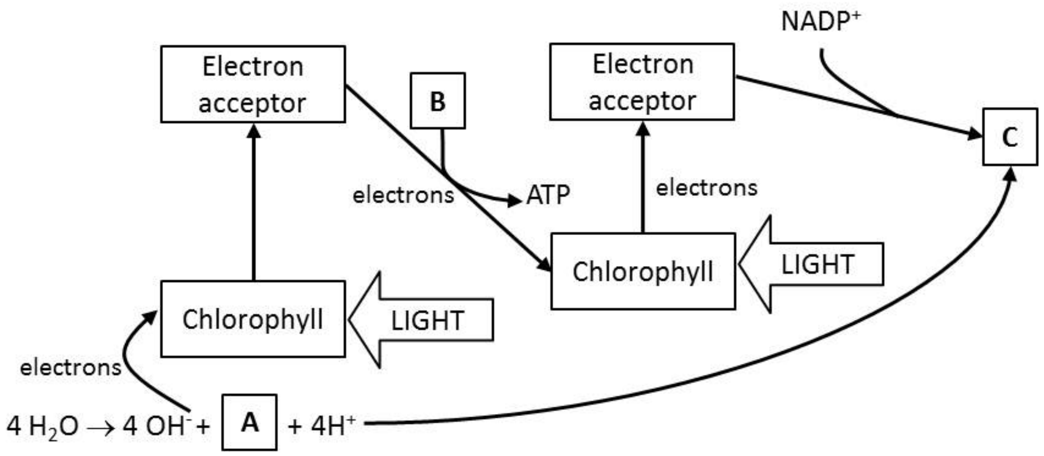

Fig. 2.1

(a) **(i)** Name the compounds A – C.

A ***oxygen***

B ***Adenosine diphosphate***

C ***Reduced nicotinamide adenine dinucleotide phosphate***

[1]

(ii) Explain the role of phospholipids in the synthesis of ATP.

1. Phospholipids make up lipid bilayer of thylakoid memb in chloroplasts, which is impermeable to H+ ions due to hydrophobic core made up of hydrocarbon tails of fatty acid chains;;

2. allow proton gradient to be set up across thylakoid memb which is used for chemiosmotic synthesis of ATP;; [2]

(b) With reference to Fig. 2.1, explain the role of water in photosynthesis.

1. Water acts as primary electron donor of non-cyclic photophosphorylation;;

2. Replenishing the electron hole/ deficit in photosystem II when photon of light has caused photoexcitation of electron from chlorophyll pigment/ loss of electron to primary electron acceptor;; [2]

For Examiner's Use

Scientists investigated the effects of temperature on the rate of respiration and photosynthesis in creeping azalea. Fig. 2.2 shows the results of temperature on the net rate of photosynthesis at three different light intensities.

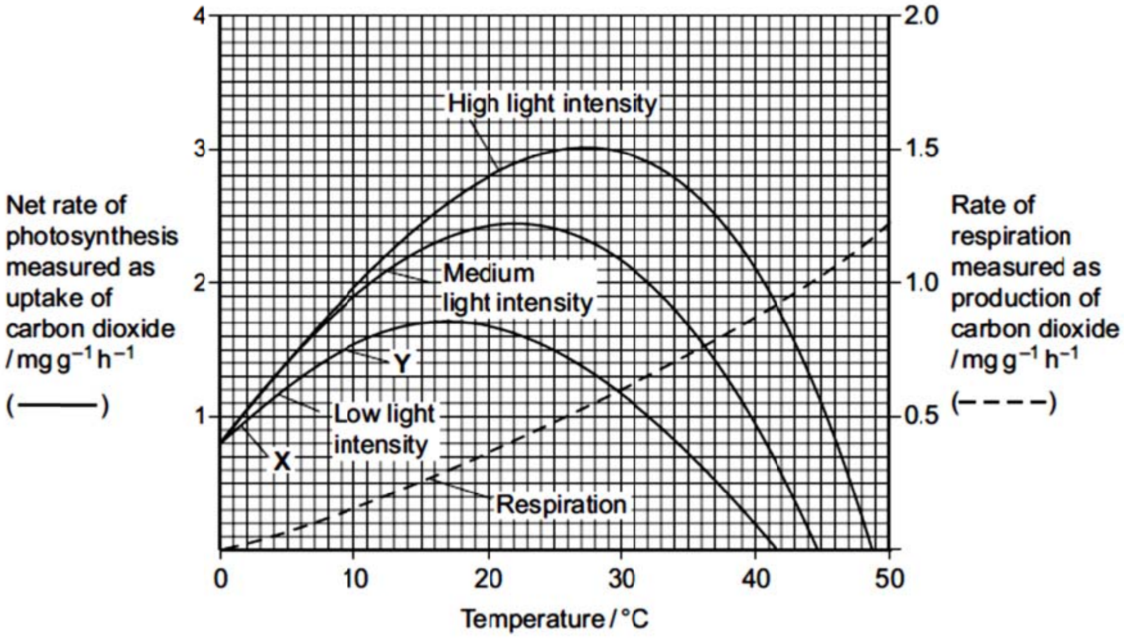

Fig. 2.2

(c) **(i)** Name the factors that limited the rate of photosynthesis between X and Y.

Light intensity and temperature;;

[1]

(ii) Explain your answers in **c(i)**.

1. When light intensity increased from "low" to "medium", net rate of P increased;;

2. When temperature increased from 2°C at X to 10°C at Y, net rate of P increased from 0.9mg g-1 h-1 to 1.5mg g-1 h-1;;

[2]

(d) Creeping azalea is a plant which grows on mountains. Scientists predict that in the area where this plant grows the mean summer temperature is likely to rise from 20°C to 23°C. It is also likely to become much cloudier.

Suggest how these changes are likely to affect the growth of creeping azalea.

1. Effect of increase in temp from 20°C to 23°C results in small increase in net P rate, but decrease in light intensity due to cloud cover results in a more significant decrease in net P rate;;

2. Net P rate likely to decrease → rate of glucose produced lower

[2]

→ lower rate of increase in dry mass → slower plant growth;;

[Total: 10]

3 The Rhesus blood group is genetically controlled. The gene for the Rhesus blood group is found on chromosome 1 and has two alleles. The allele for Rhesus positive, **R**, is dominant to that for Rhesus negative, **r**.

Another gene coding for the ABO blood groups is found on chromosome 19 and both genes play an important role in the determination of suitable donors for blood transfusion.

Fig. 3 shows the inheritance of the Rhesus blood group in one family.

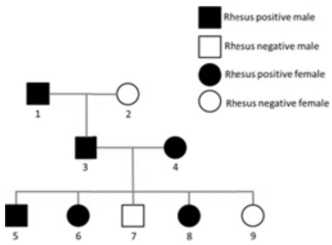

Fig. 3

(a) Explain the term *dominant*.

1. Dominant allele will be expressed and its effects will mask the effects of the recessive allele;;

2. Both homozygotes and heterozygotes show same phenotype/ dominant allele will be expressed in both homozygous and heterozygous conditions;; [2]

(b) State **one** piece of evidence from Fig. 3 that shows **R** is the dominant allele in the determination of the Rhesus phenotype.

1. Rh negative individuals 7 and 9 resulted from Rh positive individuals 3 and 4 → shows that 3 and 4 are heterozygotes;; (which possess the genotype Rr, hence individuals 7 and 9 had genotypes rr) [1]

For Examiner's Use

(c) Individual **9** with blood group B married a male with blood group A who is Rhesus positive. The couple had a daughter who is Rhesus positive with the blood group AB and a son who is Rhesus negative with the blood group O.

Using appropriate symbols, illustrate the information in a genetic diagram.

Cross between rr I^BI^O x Rr I^AI^O

1. ***Legend for blood groups + Punnet Square;;***
2. ***Correct formation of gametes from ♀ parent;;***
3. ***Correct formation of gametes from ♂ parent;;***
4. ***Correct genotypes coupled with correct phenotype;;***

® Gametes not circled = -1mark

[4]

(d) State the probability that the couple's next child will be a boy who is Rhesus negative with the blood group A.

Probability : ***1/8 x ½ = 1/16;;*** [1]

[Total: 8]

For Examiner's Use

4 Plasmids are often used as vectors in genetic engineering.

(a) **(i)** Explain what is meant by a *vector*.

1. DNA molecule/ molecular carrier into which fragments of DNA may be inserted prior to introduction into host cell;; [1]

(ii) Describe the role of restriction endonucleases in the formation of recombinant plasmids.

1. REs cleave DNA at restriction sites, generating blunt/ sticky ends;;

2. Allows DNA molecules cut by same RE to be annealed together in the presence of ligase;; [2]

(b) Fig. 4.1 shows the position of four unique restriction sites, J, K, L and M on a plasmid.

Gel electrophoresis is carried out to separate the restriction fragments after treatment with endonucleases. The results of electrophoresis are shown in Fig. 4.2.

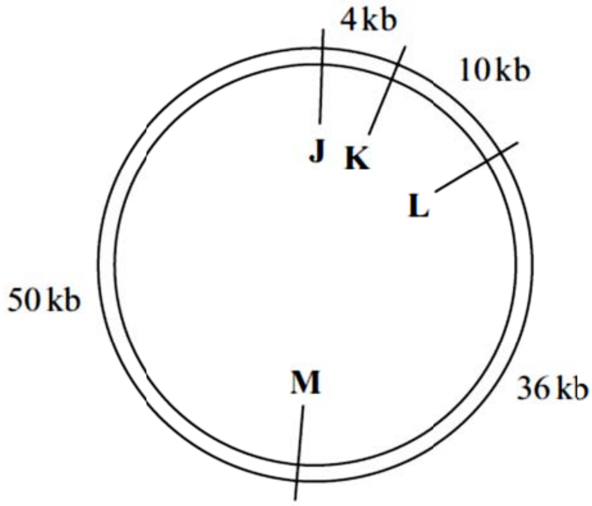

Fig. 4.1

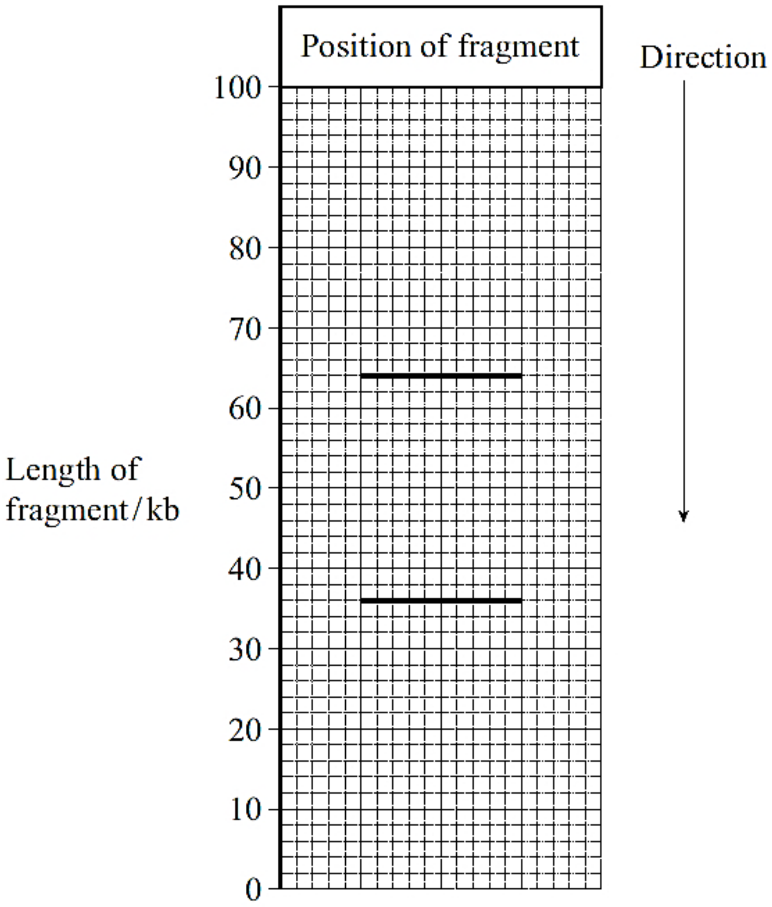

Fig. 4.2

With reference to Fig. 4.1 and 4.2,

(i) state which restriction sites were cleaved.

M and L;; [1]

(ii) Explain how the restriction fragments were separated by gel electrophoresis.

1. ***restriction fragments loaded in wells of agarose gel near cathode / negative electrode, gel immersed in TBE buffer to establish appropriate pH & provide ions for conductivity;;***
2. ***DC / voltage/ electrical field applied across the gel allowing -vely charged DNA fragments (due to presence of PO4 grp) to migrate towards +vely charged anode;;***
3. ***agarose gel act as molecular sieve impeding movement of large fragments → distance migrated inversely proportional to fragment size;***

[3]

(iii) Restriction site M is found within the *lac* Z gene and restriction site L is found within the tetracyclin resistant gene.

For Examiner's Use

Suggest why restriction enzyme L is unsuitable to be used in the formation of a recombinant plasmid.

1. ***Using RE(L) results in cleaving of tetracyclin resistant gene in plasmid → foreign gene inserted leads to disruption to tetracyclin resistant gene;;***

2. ***Unable to screen for transformed bacteria cells with recombinant plasmids as due to inability to synthesize proteins involved in tetracyclin resistance → sensitive to tetracyclin/ die when exposed to tetracyclin;;*** [2]

[Total: 9]

Section B

Answer **one** question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

5 **(a)** Describe the neo-Darwinism theory of evolution. [6]

1. ***All organisms are capable of producing large number of offspring than can survive**, but size of most populations **remains relatively constant** as many of the offspring die before they reach reproductive age due to limited resources and predators;;*

2. *There exists a **struggle for existence** where individuals of a species are constantly **competing with each other for limited resources** (e.g. food, mates and shelter) and in this struggle for existence, only a few individuals reach maturity & reproduce;;*

3. *There exists **variation among individuals within a population** which arise spontaneously, and many of these traits/characteristics are **heritable**;;*

4. *Natural selection determines **survival of the fittest**/ individuals better adapted to the environment are selected for → **able to survive to reproductive age and produce viable, fertile offspring**;;*

5. *Like produces like when individuals who survive long enough to breed successfully are likely to **produce offspring similar to them**, **passing on genes coding for advantageous characteristics to the offspring**;;*

6. ***Formation of a new species** occurs when **genetic isolation** (i.e. no gene flow) occurs/ **reproductive barriers** set sin between 2 diversifying subpopulations of individuals;;*

(b) Explain with relevant examples, how anatomical, embryological and molecular homology supports natural selection. [8]

1. *Homology refers to similarity in characteristics resulting from shared ancestry even though they may have different functions;;*
2. *Anatomical homology refers to organisms having morphological structures such as bone, organs and structural features derived from common ancestor;;*
3. *supports evolutionary relatedness because closely related species share similar features;;*
4. *E.g. Flipper of dolphin, forelimb of human, wings of bat, are homologous structures;;*
5. *Vestigial structures are anatomical structures with reduced/ no functions but resemble functional structures of other vertebrates;;*
6. *suggesting that these structures were derived from common ancestor eg. Tail bone/ appendix in humans;;*
7. *Embryological homology refers to organisms undergoing similar stages/ pathways during embryonic development shared anatomical characteristics in embryos indicate common ancestry;;*

8. *e.g. all vertebrate embryos have gill pouches on sides of throat;;*
9. *Molecular homology refers to similarities of the DNA and amino acid sequences between organisms indicating degree of relatedness between organisms/ common ancestry;;*
10. *more distantly related species will exhibit greater number of evolutionary differences than those closely related;;*
11. *eg. amino acid sequences of cytochrome C / haemoglobin;;*

(MAX 8)

(c) Explain the ways in which islands favour speciation. [6]

1. *Islands posed geographical barriers/ separated by large bodies of water → Prevent gene flow as interbreeding between the organisms is not possible → allopatric speciation;;*
2. *Sympatric speciation occurs as a result of diff conditions within island;*
3. *Natural variation exists in the population of organism, and those with adaptations to the unique selection pressure on separated islands are at selective advantage;;*
4. *Higher survival and reproductive success, Genes for traits adapted to the selection pressure on each island passed on to offsprings;;*
5. *Accumulation of these inheritable conditions may be in the areas of reproduction → Results in pre- and post-zygotic barriers;;*
6. *Speciation occurs when the individuals in 2 populations are not able to mate to produce viable, fertile offsprings → speciation occured as a result of adaptive radiation;;*
7. *Spontaneous mutations created new alleles in separated populations → genetic divergence between separated populations;;*
8. *Founder's effect → small gene pool with rare alleles which are selected for and propagated → genetic divergence from original population;;*

(MAX 6)

[Total: 20]

6 **(a)** Explain the normal functions of stem cells in a living organism. [6]

1. *unlimited renewal capacity, cont division to replace worn out tissues;;*
2. *ESCs, derived from embryos are pluripotent;;*
3. *mature/ differentiate into any cell type except extra-embryonic tissue/ placenta;;*
4. *Mature and differentiate into the 3 layers, ectoderm, mesoderm and endoderm;;*
5. *HSCs, (found in the bone marrow) multipotent;;*
6. *can mature/ differentiate and give rise to myeloid and lymphoid progenitor cells;;*
7. *that can differentiate into the different matured blood cell types where myeloid progenitor cells give rise to RBCs, WBCs and platelets while lymphoid progenitor cells give rise to lymphocytes;;*

Award 1m for mention of other normal functions of stem cells e.g. totipotent

zygotic stem cells

(b) Discuss the goals and implications of the human genome project, including the benefits and difficult ethical concerns for humans. [8]

Goals of HGP

1. ***Gene identification of all the genes in human DNA for construction of genomic maps and study of the sequence and function of the protein it codes for;;***
2. ***Sequence determination of the nucleotides that make up human DNA;;***
3. ***Storage of this information in databases to facilitate research in both private and government sectors worldwide;;***
4. ***Improvement of tools for data analysis → enable the development of DNA clone libraries representing the individual chromosomes;;***

(max 2)

Benefits of HGP

5. ***Improved disease diagnosis because HGP provides knowledge on gene sequences of disease-causing alleles;;***
6. ***Earlier detection of genetic predispositions to disease by identifying causative mutations to be used as disease markers;;***
7. ***Rational drug design by improving disease treatment with specific drug design to better target diseased cell/ informs personalized medicine;;***
8. ***Aid in gene therapy since linkage and physical maps have made it easier for researchers to locate genes to study their functions/ gene interaction in complex genetic diseases;;***

(max 3)

Ethical concerns of HGP

9. ***Fairness in use of genetic information → use of genetic information disclosed to various bodies should be fair to individual and all concerned parties;;***
10. ***Privacy and confidentiality of genetic information → individual privacy and confidentiality should be maintained, and that an individual's genetic information is not distributed to other parties without permission;;***
11. ***Genetic discrimination → by insurance companies/ employers/ stigmatization by society;;***
12. ***Adequate informed consent about complex and controversial procedures before decision making;;***
13. ***Intellectual property rights → patenting gene fragments potentially hinders research and innovation of new technology, and promotes monopolisation of gene diagnostic market by biotechnology firms;;***

(max 3)

(MAX 8)

(c) Comment on the statement "GM crops can make a relevant contribution to solving health problems in developing countries." [6]

1. ***state e.g. Golden Rice, which's a genetically modified, yellow-orange rice grain that contains β-carotene, a precursor to vitamin A;;***
2. ***treat vitamin A deficiency in children in In developing countries, due to it being a leading cause of vision impairment and blindness & infectious diseases;;***
3. ***For :***
    a. ***Rice is the staple diet, as compared to other Vit A sources e.g. green leady vegetables which are expensive and seasonal;;***
    b. ***Rice can be consumed in larger amount per serving as compared to the other sources which require more than one servings to ensure adequate Vit A intake;;***
    c. ***It is a sustainable and cost-effective alternative to vitamin supplements;;***
    d. ***Rice, with enhanced micronutrient content becomes more nutritious and may help reduce malnutrition in poor countries.***
4. ***Agst:***
    a. ***Malnourished children w/o enough fat stores, hence not able to absorb β-carotene which's fat soluble;;***
    b. ***key to overcoming micronutrient deficiencies more likely lies in improved and varied diets and improved socioeconomic status;;***
    c. ***Increased levels of β-carotene may have impact on organisms downstream of the food chain or may lead to ecological damage to soil organisms which rely upon rice for existence;;***
    d. ***Rice eaten in large quantities, could lead to excessive intake of vitamin A. leading to hypervitaminosis/ vitamin A toxicity resulting in developmental problems in infants.***

(MAX 6)

[Total: 20]